

Session 2 : AI Terminology and Machine Learning

Session-2 Objectives

By the end of this session, you will be able to...



**Why Terminology
Matters**



**Functional Areas
within AI & ML**



**Machine
Learning**



**Types of Machine
Learning**

Why Terminology Matters

Before we build — let's speak the same language



Jargon kills adoption

Teams disengage when AI feels too technical. Words like 'stochastic gradient descent' create fear, not understanding.



Confusion costs decisions

Leaders make poor AI investment decisions when they can't evaluate vendor claims or understand what a model actually does.



Clarity unlocks power

When you know what '**training data**', '**inference**' and '**parameter**' mean, you can direct AI tools — not just use them.

The jargon barrier stops 80% of people before they start.

Term 1 of 8 — Algorithm

The recipe that tells AI what to do with data



ALGORITHM

A set of step-by-step rules or instructions that a computer follows to solve a problem or make a decision.



Analogy

A recipe is an algorithm for cooking. It tells you exactly what to do, in what order, with what ingredients — and produces a consistent result every time.

Business example: The algorithm that decides which emails go to your spam folder — it checks sender, subject, and content patterns.

1

INPUT

Receive raw data
(email text, pixels, numbers)

2

PROCESS

Apply rules & calculations
(check patterns, compute scores)

3

DECISION

Compare result to threshold
(is score > 0.7?)

4

OUTPUT

Return prediction or action
(spam / not spam)

Term 2 of 8 — Training Data

The experience AI learns from — just like humans learn from examples



TRAINING DATA

The labelled examples that an AI model learns from. The quality, quantity, and diversity of training data directly determines how well the model performs.



Quantity

More examples = better generalisation. GPT-4 trained on ~13 trillion tokens of text.



Quality

Garbage in, garbage out. Noisy or biased data creates unreliable models.



Diversity

Data must represent all groups and scenarios the model will encounter in the real world.

Real-world examples:

"ChatGPT trained on ~45TB of text from the internet, books, and Wikipedia"

"A medical AI for X-ray diagnosis may train on 1M+ labelled scans"

"Tesla's FSD trains on billions of miles of real driving footage"

"A spam filter trains on millions of labelled spam / not-spam emails"

Term 3 of 8 — Model

The trained brain — the output of learning



MODEL

A mathematical representation of patterns learned from training data. When you run ChatGPT, Claude, or any AI tool — you are interacting with a model. It is the end-product of the AI training process.

BEFORE TRAINING

- ✗ Raw algorithm (untrained)
- ✗ No knowledge of domain
- ✗ Random, useless outputs
- ✗ Just code + math

Train →

AFTER TRAINING

- ✓ Trained model (e.g. GPT-4)
- ✓ Domain knowledge encoded
- ✓ Useful, accurate outputs
- ✓ Billions of parameters set

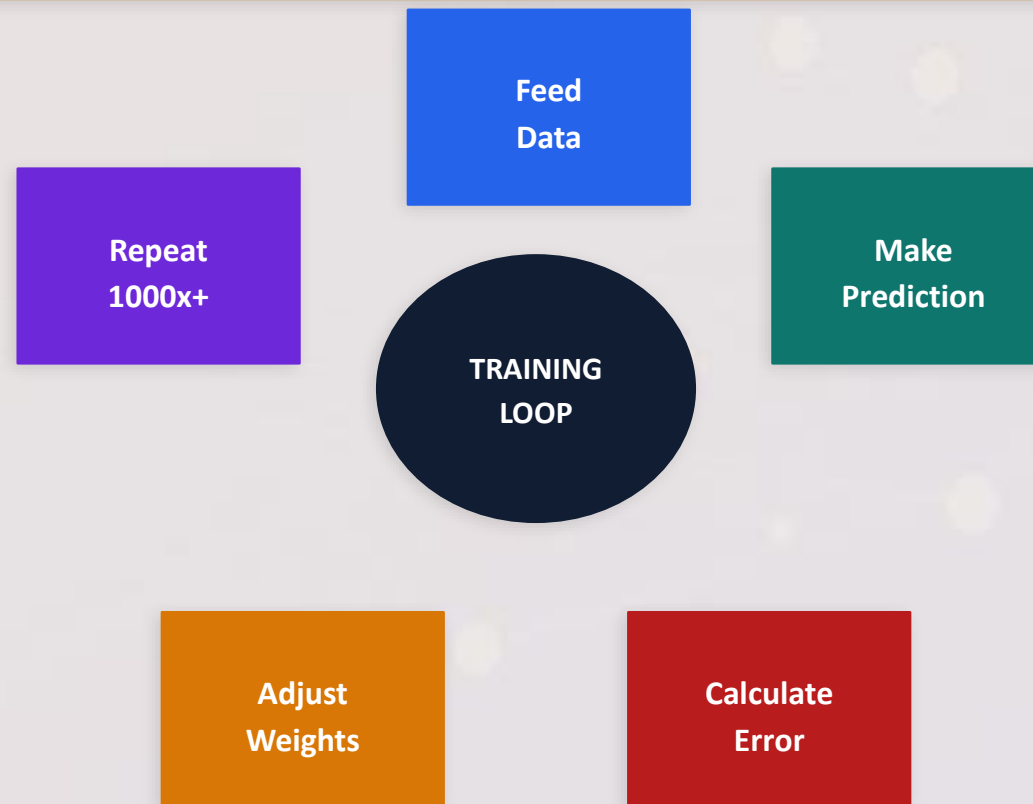
MODEL TYPES

- Language Model
- Image Model
- Speech Model
- Forecasting Model
- Recommendation Model

Term 4 of 8 — Training

How an AI learns — the iterative improvement loop

TRAINING — The process of repeatedly exposing an algorithm to training data and adjusting its internal parameters (weights) until it produces accurate predictions. Like a student practising maths: wrong answers trigger adjustments until they get it right.



🕒 GPT-4 training took weeks on thousands of specialised chips (H100 GPUs) at a cost of ~\$100 million. Your inference (using the model) takes milliseconds.

Term 5 of 8 — Parameters

The billions of dials that define what a model knows

175 Billion

parameters in GPT-3

GPT-4: estimated 1+ Trillion

Claude 3 Opus · Gemini Ultra · similar scale

Each parameter = one tiny learned number.
Together they encode the model's "knowledge".

What are parameters?

Numerical values (weights) inside the neural network. Adjusted during training to reduce prediction errors. After training, they are frozen — this is your model.

The human brain analogy

Think of parameters as synaptic connections between neurons. More connections = more capacity to learn complex patterns. The brain has ~100 trillion synapses.

Why do they matter to you?

Larger models (more parameters) are generally more capable but slower and costlier. Smaller models are faster and cheaper. Choosing the right model size matters for business deployment.

Term 6 of 8 — Inference

When the trained model goes to work — this is the moment you actually use AI



INFERENCE — The act of using a trained model to make predictions or generate outputs on new, unseen data. Every time you type a prompt into ChatGPT, Google Maps calculates a route, or Netflix suggests a film — that is inference.

TRAINING

When?:	Once — before deployment (or periodically retraining)
Cost?:	Very high — weeks, specialist hardware, millions of \$
Data?:	Massive labelled datasets required
Output?:	A trained model (frozen weights)
Who does it?:	Data science / ML engineering teams

INFERENCE

When?:	Every time a user sends a query — milliseconds
Cost?:	Low per query — scales with usage volume
Data?:	Single new input from the user
Output?:	Prediction, generation, or decision
Who does it?:	The model — automatically, instantly

Term 7 of 8 — Tokens

The atomic units of text that AI models actually read



TOKEN — A chunk of text (roughly 4 characters or $\frac{3}{4}$ of a word). AI models don't read words — they read tokens. Knowing this explains why AI has context limits and why pricing is per token.

How the sentence below gets tokenised by a language model:



← 10 tokens

1 token ≈ 4 chars

English text averages ~0.75 words per token

Context window

Max tokens a model can 'see' at once.
GPT-4: 128k tokens ≈ ~100,000 words

Pricing

Most AI APIs charge per 1,000 tokens input + output. OpenAI GPT-4: ~\$0.03/1k

Why it matters

Long documents need to fit in context window. Knowing tokens helps you manage cost and limitations.

Term 8 of 8 — Prompt

Your instruction to the model — the most powerful skill in this programme



PROMPT — The text input you give to an AI model to generate a response. The quality of your prompt is the single biggest factor in the quality of the AI output. Prompt engineering is the skill of crafting effective instructions.

The difference a good prompt makes:

✗ WEAK PROMPT		✓ STRONG PROMPT	
<i>"Write me something about marketing."</i> Result: Vague, generic content. Could be 1 sentence or 10 pages. No format. No audience. No purpose. The model guesses everything.		<i>"Act as a senior B2B marketing strategist. Write a 200-word LinkedIn post for a SaaS CFO audience about why AI reduces finance team costs. Use a hook opening, 3 bullet points, and a CTA."</i> Result: Targeted, formatted, audience-specific, actionable content. The model has everything it needs.	
Role	Task	Context	Format
'Act as a [expert]'	'Write / Analyse / ...'	Audience + purpose	Length + structure

Your AI Glossary — All 8 Terms

Algorithm

Step-by-step rules a computer follows to solve a problem

Model

Mathematical representation of learned patterns — the output of training

Parameters

Billions of numerical values inside the model that encode its 'knowledge'

Tokens

Chunks of ~4 characters that AI models process instead of whole words

Training Data

Labelled examples the AI learns patterns from

Training

Process of adjusting model parameters using data until accuracy improves

Inference

Using the trained model to make predictions on new data

Prompt

Your text instruction to the AI model — quality determines output quality

Jargon-Busting Exercise

Can you match the term to the plain-English definition?

 Instructions: For each vendor statement below, identify the AI term being used and translate it into plain English for a business stakeholder.

"Our system has been trained on 50 billion labelled examples across 47 languages"

Term: Training Data

→ They used massive, diverse labelled datasets to teach the model — expect broad language coverage and reliability.

"The model performs inference in under 200 milliseconds at scale"

Term: Inference

→ The AI produces results in under 0.2 seconds when users query it — fast, real-time responses.

"We use a 70-billion parameter transformer architecture"

Term: Parameters + Model

→ It's a large, capable AI model — expect high quality but higher compute cost.

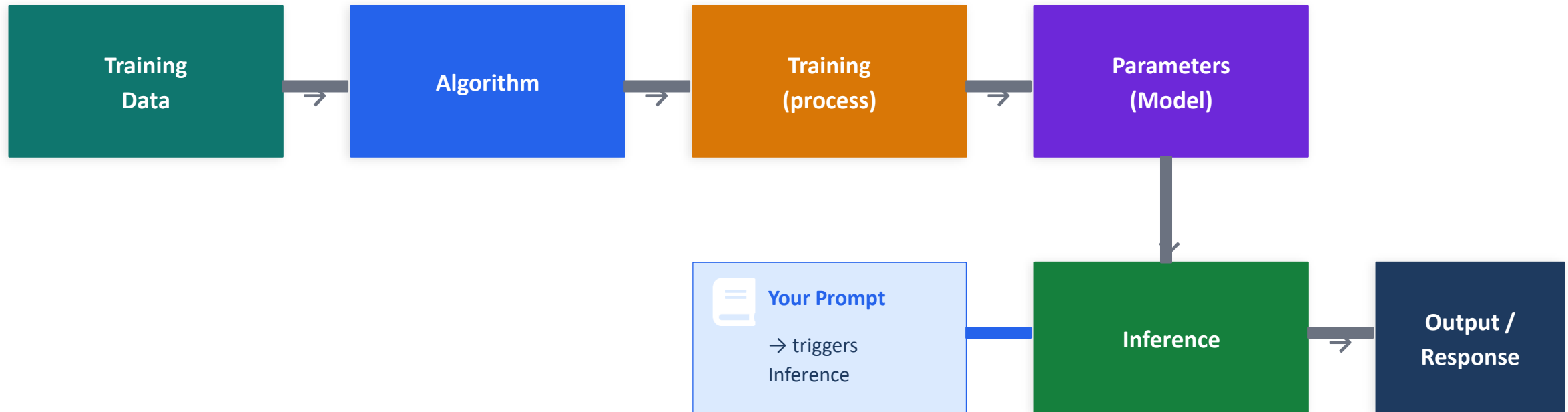
"Prompt injection attacks exploit vulnerabilities in the instruction layer"

Term: Prompt

→ Hackers try to manipulate the AI by crafting malicious input instructions — a security concern for enterprise deployments.

How the Terms Connect

The complete AI system — from data to output



Reading the diagram: Training Data + Algorithm → Training process → sets Parameters (frozen in the Model) → User sends a Prompt → triggers Inference → generates Output

AI Terms in the Wild

Reading real headlines with your new vocabulary

Read each headline. Identify which terms from today you can spot. What does it actually mean?

"OpenAI releases GPT-5 with 2 trillion parameters and 1M token context window"

Parameters • Tokens • Model

New AI model with double the learned knowledge capacity, able to process ~750,000 words at once.

"Meta's Llama 3 achieves 95% accuracy after training on 15 trillion tokens of data"

Training • Training Data • Tokens

Meta ran their learning algorithm on 15 trillion text chunks — the model now correctly answers 95% of test questions.

"New technique reduces hallucination in LLMs by grounding inference with verified documents"

Hallucination • Inference • Tokens

A method that makes AI output more accurate by connecting answers to trusted sources before generating responses.

"Anthropic's prompt injection filter blocks malicious instructions before inference"

Prompt • Inference • Model

Safety system that screens user inputs for attempts to manipulate the AI before it generates a response.



Machine Learning

Functional Areas within AI & ML

Type 1. Predictive AI

Type 2. Generative AI

Predictive AI

Forecasts future outcomes using historical data

- Sales forecasting
- Employee attrition prediction
- Demand planning
- Risk assessment

👉 **Answers:** "What will happen next?"

Generative AI

Creates new content using learned patterns

- Email & report drafting
- AI chatbots
- Learning content creation
- Product/design ideas

👉 **Answers:** "What can we create next?"

Key Difference: Predictive AI helps in **decision-making**, while Generative AI enhances **creation & productivity**.

When to use

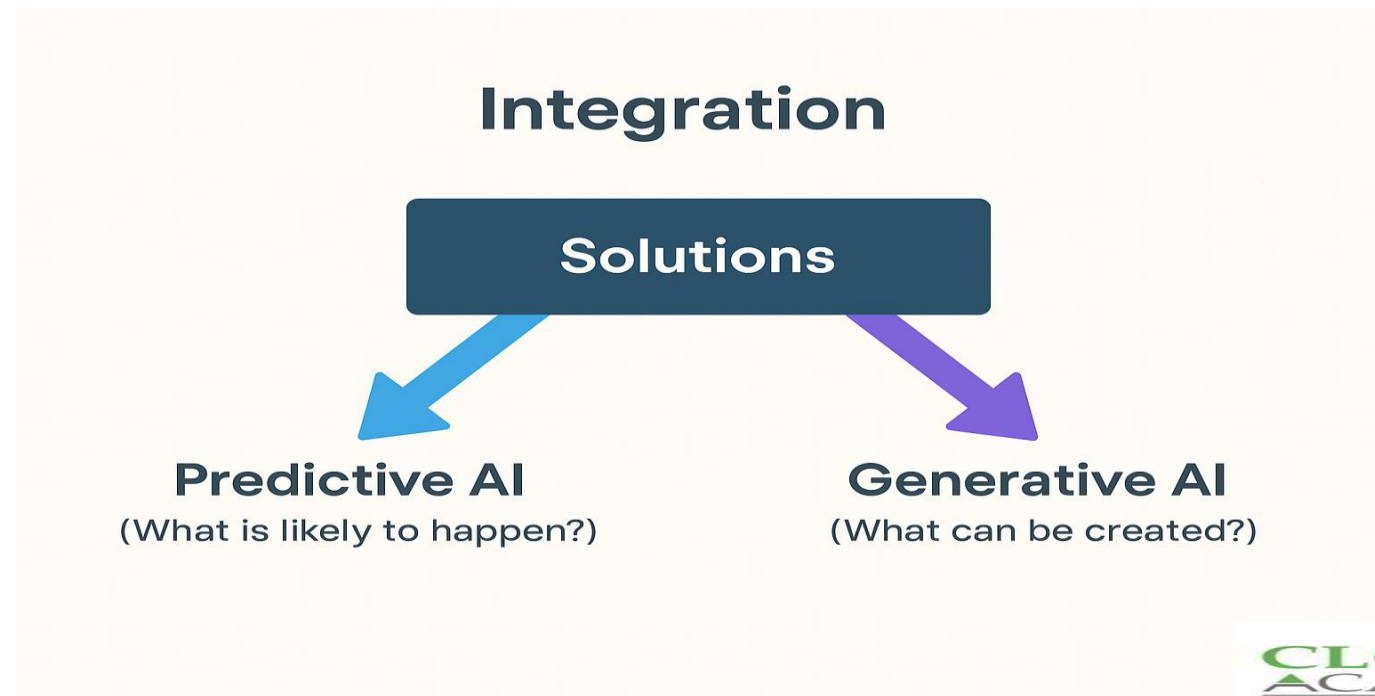
- Depends on use case

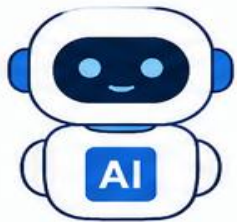
Generative AI

- Multimodal content

Predictive AI

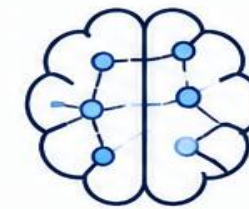
- Forecasting and Predictions





MACHINE LEARNING

Machine Learning is a subset of Artificial Intelligence that enables systems to learn from data and improve performance without being explicitly programmed.



MACHINE LEARNING

SUPERVISED LEARNING

The model learns from labeled data. It predicts output based on input-output examples.

CLASSIFICATION

Predicts a categorical (label) output.

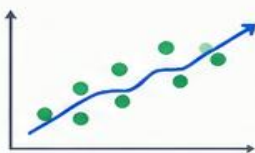


Examples:

- Email spam detection
- Image recognition
- Fraud detection

REGRESSION

Predicts a continuous (numeric) value.



Examples:

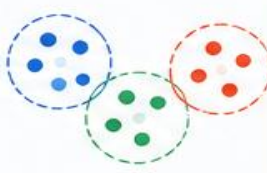
- House price prediction
- Sales forecasting
- Temperature prediction

UNSUPERVISED LEARNING

The model learns from unlabeled data. It finds patterns or structures in the data on its own.

CLUSTERING

Groups similar data points together.



Examples:

- Customer segmentation
- Document grouping
- Image clustering

ASSOCIATION RULE LEARNING

Finds relationships or associations between variables in data.



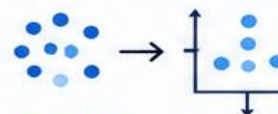
{Bread, Butter} → {Jam}

Examples:

- Market basket analysis
- Cross-selling products
- Recommendation systems

DIMENSIONALITY REDUCTION

Reduces the number of features while preserving important information.

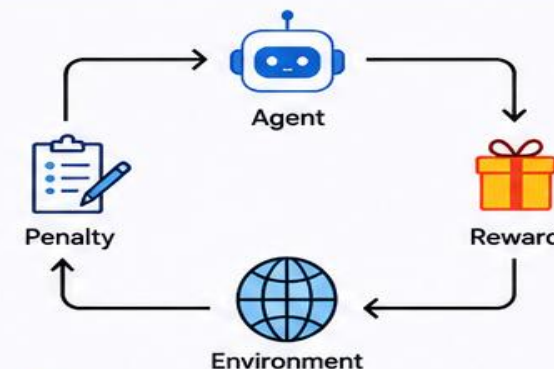


Examples:

- Feature selection
- Data visualization
- Noise reduction

REINFORCEMENT LEARNING

The model learns by interacting with an environment and receiving rewards or penalties.



Examples:

- Game playing (e.g., Chess, Go)
- Robot navigation
- Autonomous driving

HOW IT WORKS



Data Collection



Data Preparation



Model Training



Evaluation



Prediction / Decision

KEY TAKEAWAY

Machine Learning enables computers to learn from data, identify patterns, make predictions, and improve automatically across different types of problems.



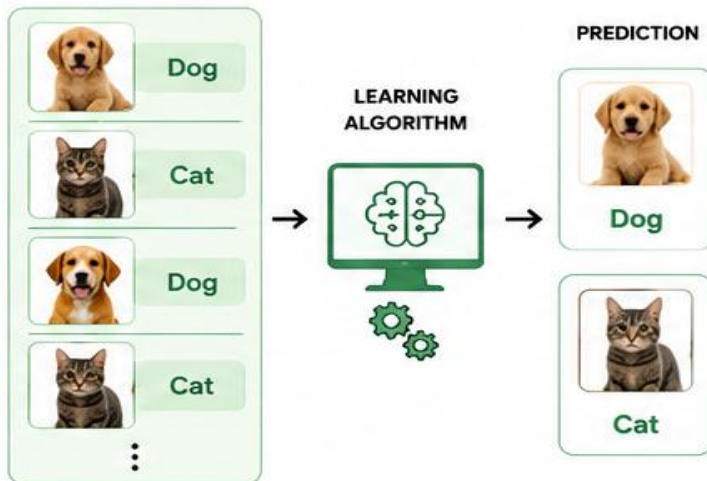
Supervised Learning



SUPERVISED LEARNING

Supervised Learning is a type of machine learning where the model learns from labeled data. It maps inputs to outputs and makes predictions on new, unseen data.

LABELLED TRAINING DATA



KEY POINTS

- ✓ Uses labeled data (input-output pairs)
- ✓ Learns the relationship between input and output
- ✓ Can predict outputs for new data

EXAMPLES



Email Spam Detection



House Price Prediction



Disease Diagnosis

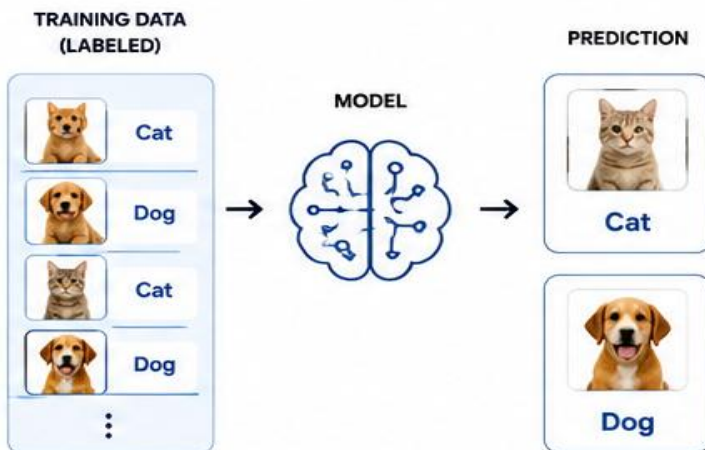


Sales Forecasting

CLASSIFICATION

Classification is a supervised learning technique used to predict a categorical (discrete) output. The output belongs to one of the predefined classes.

EXAMPLE: CAT OR DOG CLASSIFICATION



KEY POINTS

- ✓ Predicts categorical labels
- ✓ Examples of classes: Yes/No, True/False, Cat/Dog
- ✓ Common algorithms: Logistic Regression, Decision Trees, SVM, Naïve Bayes

EXAMPLES



Spam / Not Spam



Sentiment Analysis



Customer Churn
Yes / No

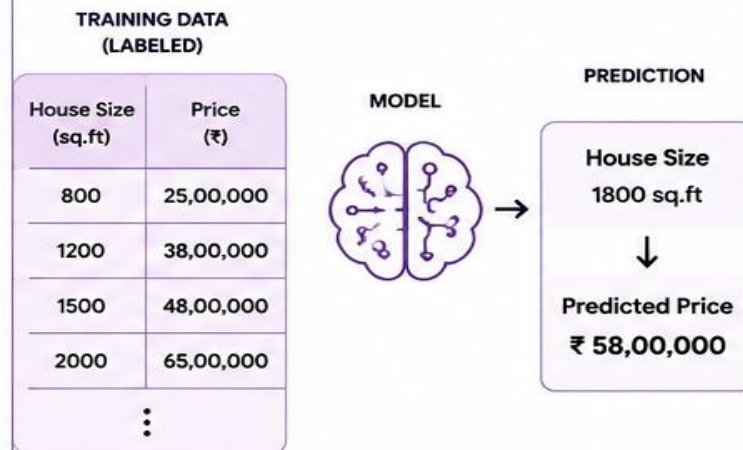


Fraud Detection

REGRESSION

Regression is a supervised learning technique used to predict a continuous (numeric) output. The output can take any real value.

EXAMPLE: HOUSE PRICE PREDICTION



KEY POINTS

- ✓ Predicts continuous numeric values
- ✓ Examples: Price, Temperature, Salary, Sales
- ✓ Common algorithms: Linear Regression, Ridge Regression, Random Forest Regression

EXAMPLES



House Price Prediction



Sales Forecasting



Temperature Prediction



Salary Prediction

Question 1: What type of data is used in supervised learning?

A. Unlabeled data

B. Random data

C. Labeled data

D. Encrypted data

Answer: C. Labeled Data

Question 2: Which supervised learning technique predicts categories?

A. Regression

B. Classification

C. Clustering

D. Reinforcement Learning

Answer: B. Classification

Question 3: Regression is mainly used to predict:

A. Images

B. Text Labels

C. Categories

D. Continuous Numerical Values

Answer: D. Continuous numerical values

Question 4: Which of the following is an example of classification?

A. Predicting house prices

B. Forecasting sales

C. Spam email detection

D. Temperature prediction

Answer: C. Spam email detection



UNSUPERVISED LEARNING



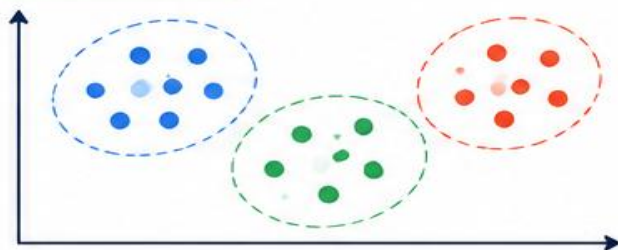
Unsupervised Learning is a type of machine learning where the model learns from unlabeled data. It discovers hidden patterns, structures, or relationships in the data on its own.

TYPES OF UNSUPERVISED LEARNING

1 CLUSTERING

Clustering groups similar data points together based on their patterns or characteristics.

EXAMPLE



COMMON ALGORITHMS

- K-Means Clustering
- Hierarchical Clustering
- DBSCAN
- Gaussian Mixture Models (GMM)



APPLICATIONS



Customer Segmentation



Market Basket Analysis



Document Clustering



Social Network Analysis

2 ASSOCIATION RULE LEARNING

Association finds interesting relationships or associations between variables in large datasets.

EXAMPLE



IF

{Bread, Butter}

THEN

→ {Jam}

(People who buy Bread and Butter are likely to buy Jam.)

COMMON ALGORITHMS

- Apriori Algorithm
- Eclat Algorithm
- FP-Growth Algorithm



APPLICATIONS



Market Basket Analysis



Cross-selling Products



Recommendation Systems

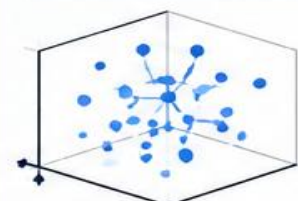


Customer Behavior Analysis

3 DIMENSIONALITY REDUCTION

Dimensionality Reduction reduces the number of features while preserving important information.

EXAMPLE



High Dimensional Data (3D)



Reduced Dimensional Data (2D)

COMMON ALGORITHMS

- Principal Component Analysis (PCA)
- Singular Value Decomposition (SVD)
- t-SNE
- Autoencoders



APPLICATIONS



Data Visualization



Improved Model Performance



Reducing Storage



Noise Reduction



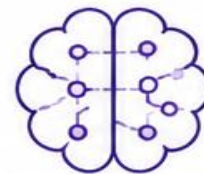
KEY TAKEAWAY:

Unsupervised Learning helps in exploring data, finding hidden patterns, and extracting meaningful insights without any predefined labels.



REINFORCEMENT LEARNING

Reinforcement Learning is a type of machine learning where an agent learns by interacting with an environment to achieve a goal and maximize cumulative reward.



KEY IDEAS



An **agent** takes actions in an environment.



The **environment** responds with a new state and a **reward** (or penalty).

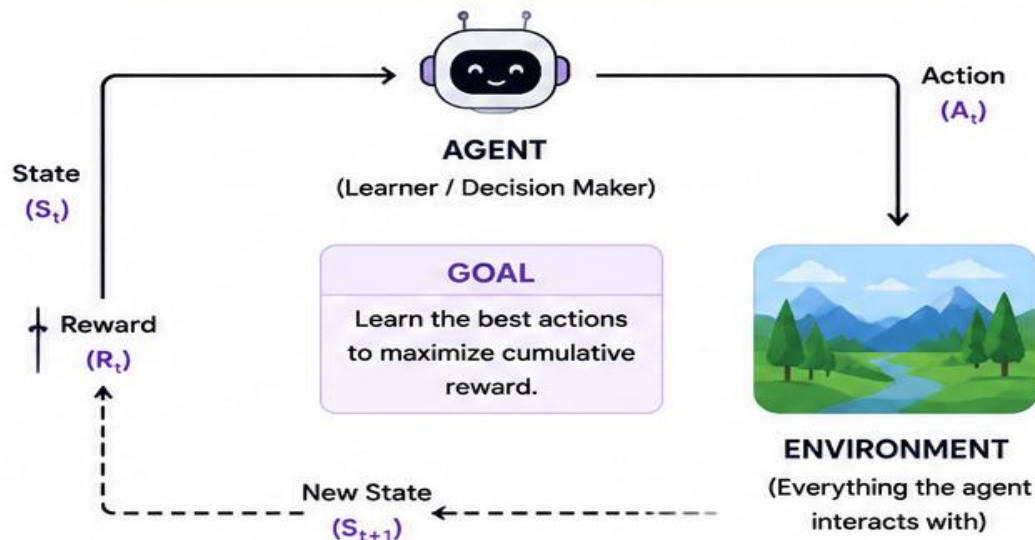


The agent learns the best strategy (**policy**) to maximize total reward over time.



Learning improves through **trial** and **error**.

HOW REINFORCEMENT LEARNING WORKS



COMPONENTS



Agent: Learns and makes decisions.



Environment: External world the agent interacts with.



Action: Choices made by the agent.



Reward: Feedback from the environment (positive or negative).



Policy: Strategy that defines the agent's behavior.

COMMON ALGORITHMS

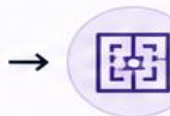
- Q-Learning
- Deep Q-Networks (DQN)
- Policy Gradient
- Actor-Critic Methods
- Proximal Policy Optimization (PPO)
- Monte Carlo Tree Search (MCTS)



TYPICAL PROCESS



Define Goal



Agent Takes Action



Environment Responds



Reward / Penalty



Agent Learns and Improves

APPLICATIONS



Game Playing



Robotics



Autonomous Driving



Resource Management



Trading



Recommendations



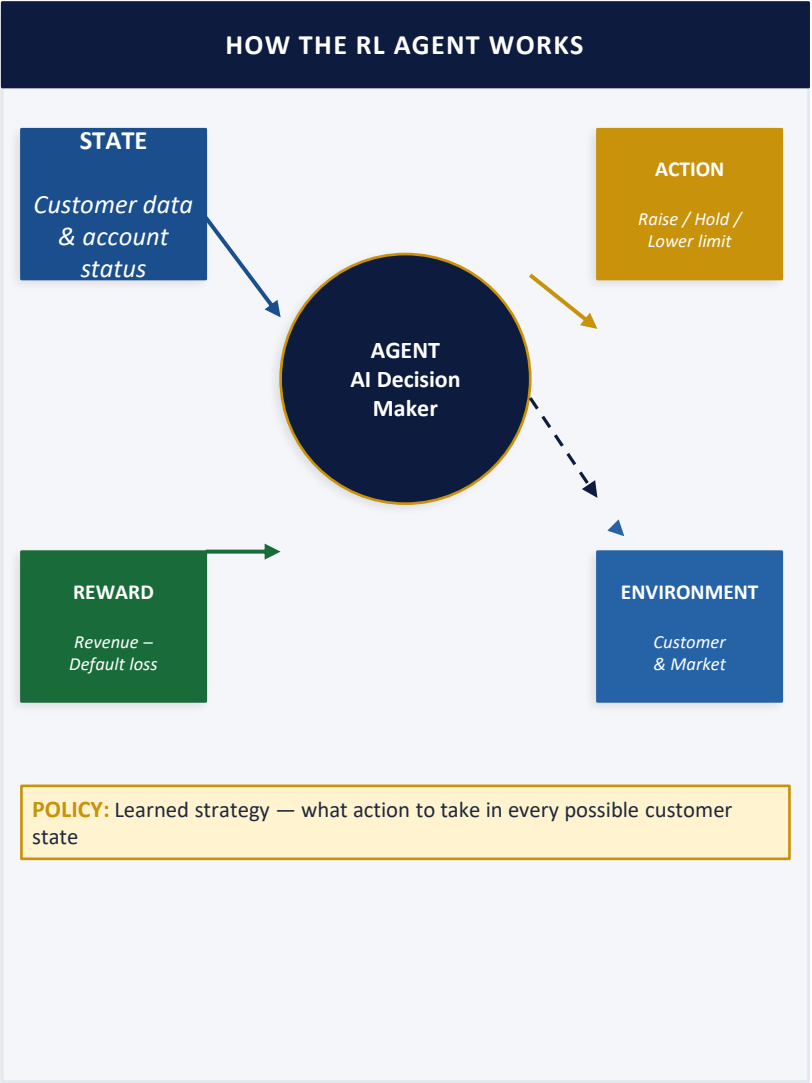
KEY TAKEAWAY: Reinforcement Learning enables agents to learn optimal behavior through interaction, feedback, and continuous improvement to achieve long-term goals.





The Challenge: A bank must decide in real-time whether to raise, hold, or lower a customer's credit limit — balancing **revenue**, **retention**, and **credit risk** across millions of accounts simultaneously.

① RL MECHANISM



② LIVE DECISION LOOP



③ BANK PERFORMANCE IMPACT

MEASURABLE BANK OUTCOMES		
BEFORE — Static Rules	AFTER — RL Agent	
Credit Loss Rate		
✗ 2.8% of portfolio	✓ 1.6% of portfolio	
Revenue per Account		
✗ £148 / yr avg	✓ £194 / yr avg	
Limit Review Cycle		
✗ Quarterly (manual)	✓ Real-time (automated)	
False Declines		
✗ ~22% of requests	✓ ~9% of requests	
Customer Complaints		
✗ High (blunt rules)	✓ Low (personalised)	
43% reduction in credit losses	31% increase in interest revenue	Real-time decisions vs quarterly cycle
Key Insight: The RL agent has now 'experienced' more credit decisions in simulation than a team of 100 analysts could make in a lifetime — and its policy continuously improves with live data.		

Which statement correctly distinguishes **TRAINING** from **INFERENCE** in a machine learning system?

A

Training is fast and cheap; Inference is slow and expensive

B

Training and Inference both happen in real-time when a user sends a query

C

Training happens once using massive data to build a model; Inference uses that model to answer new queries in milliseconds

D

Training requires user input; Inference requires labelled data

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**Training happens once using massive data to build a model;
Inference uses that model to answer new queries in
milliseconds**

✓

D

Training requires user input; Inference requires labelled data

CORRECT ANSWER: C Training is the expensive, one-time process (GPT-4 cost ~\$100M and took weeks). Inference is instant and cheap — every time you type a prompt or a fraud model scores a transaction, that is inference. The key asymmetry: training is done by ML teams; inference is done by the model for every user query.

A bank wants to predict NEXT QUARTER'S REVENUE from current transaction volumes. Which supervised learning technique should they use?

A

Classification — it predicts a category (high/medium/low revenue)

B

Regression — it predicts a continuous numerical value (the exact revenue figure)

C

Clustering — it groups similar revenue periods together without labels

D

Reinforcement Learning — it optimises revenue allocation decisions

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D

Reinforcement Learning — it optimises revenue allocation decisions

CORRECT ANSWER: B Regression predicts a continuous number — like £4.7M in revenue. The key test: if the output is a number on a scale, it's Regression. If it's a category label (fraud/not fraud, approve/decline), it's Classification. Revenue is always a number, so Regression is the right choice.

A bank feeds 3 years of customer transaction data into an ML model — with NO predefined categories — and asks it to find natural groupings. What type of ML is this?

A

Supervised Learning — Classification, because it groups customers into segments

B

Supervised Learning — Regression, because it analyses transaction amounts

C

Unsupervised Learning — Clustering, because it finds patterns in unlabelled data

D

Reinforcement Learning, because the model learns from customer feedback rewards

3

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C

Unsupervised Learning — Clustering, because it finds patterns in unlabelled data



D

Reinforcement Learning, because the model learns from customer feedback rewards

CORRECT ANSWER: C The giveaway is 'NO predefined categories' and 'natural groupings'. Clustering is the unsupervised technique that discovers structure in unlabelled data on its own. The bank doesn't define segments in advance — the model finds them. This is how banks build customer microsegments for personalised marketing without prior assumptions.

In a Reinforcement Learning system for dynamic credit limit management, which of the following correctly maps the RL components?

A

Agent = Customer | Environment = Bank | Reward = Credit Score | Action = Spending behaviour

B

Agent = RL model | Environment = Customer & Market | Reward = Revenue minus default losses | Action = Raise/Hold/Lower the limit

C

Agent = Compliance team | Environment = Regulatory framework | Reward = No fines | Action = Writing policies

D

Agent = Training data | Environment = Neural network | Reward = Low error rate | Action = Parameter update

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✓

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D

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Reward = Low error rate | Action = Parameter update

CORRECT ANSWER: B In this RL system: the AI model is the Agent making decisions. The Customer and Market are the Environment it interacts with. The Actions are the credit limit decisions (Raise/Hold/Lower). The Reward is the business outcome — interest revenue earned minus any default losses incurred. Option D describes the training process, not an RL deployment.

A bank's compliance team wants to detect money laundering by identifying sequences of transactions that match known layering typologies — across multiple accounts and time periods. Which ML approach is MOST appropriate?

A

Supervised Learning — Regression (predicts the monetary value of suspicious transactions)

B

Supervised Learning — Classification (labels transactions as suspicious/not suspicious using historical AML cases)

C

Unsupervised Learning — Association Rule Learning (discovers transaction sequence patterns that co-occur with money laundering behaviours)

D

Reinforcement Learning (trains an agent to maximise the number of SARs filed)

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✓

D

Reinforcement Learning (trains an agent to maximise the number of SARs filed)

CORRECT ANSWER: C The key requirement is detecting unknown sequences across multiple accounts — not classifying known cases. Association Rule Learning discovers co-occurring patterns without predefined categories. B is valid for known typologies, but C finds NEW patterns the compliance team hasn't seen before. AML typologies evolve constantly — unsupervised discovery is what catches novel schemes before human analysts recognise them.

Session 2-3 — Key Takeaways

What you now know that most people don't

1

AI runs on algorithms, data, and parameters

Every AI system uses a learning algorithm, trained on labelled data, that adjusts billions of parameters to produce a useful model.

2

Training ≠ Inference — they are opposite ends

Training happens once (expensive, slow). Inference happens every query (fast, cheap). You interact only with inference.

3

Tokens and prompts are your control levers

Understanding tokens explains cost and limits. Mastering prompts is the highest-leverage skill for every AI user in this room.

4

Types of Machine Learning

Supervised, Unsupervised, Reinforcement Learning